

Homework 3

COMS 3310

Due Friday Feb 27th at 11:59pm

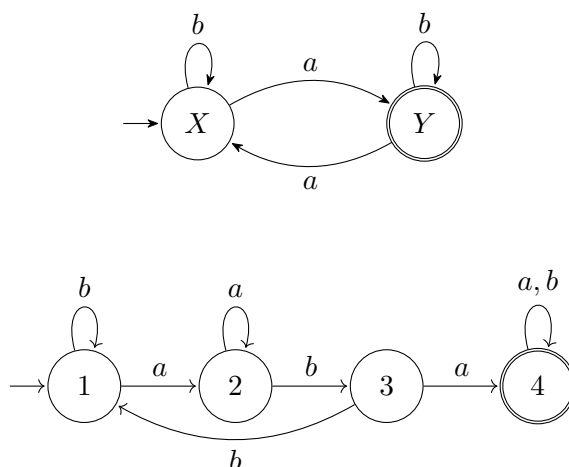
No Late Submissions !!!

Please read the HW guidelines (see next page). Failure to follow the guidelines will result in a lower grade.

Notation: $\mathbb{N}$ is the set of natural numbers, i.e.,

$$\mathbb{N} = \{0, 1, 2, \dots\}.$$

1. Suppose you are given the following two DFAs:



- (a) Design a “cross-product” DFA to accept the union of the languages of the two machines. Let your “rows” correspond to the states X and Y .
 - (b) How would your DFA change if you wanted to accept the language consisting of strings which are in both of the languages?
 - (c) How would your DFA change if you wanted to accept the language consisting of strings which are in exactly one of the two languages?
2. Given an NFA $N = (Q, \Sigma, \delta, q_0, F)$ suppose that you make all the accept states non-accept, and make all the non-accept states accept states. That is, consider $N' = (Q, \Sigma, \delta, q_0, Q \setminus F)$. Either prove that $L(N')$ is the complement of $L(N)$ or provide a counterexample.

3. Design an NFA for each of the following languages over $\Sigma = \{a, b\}$ with at most the indicated number of states.
- The language L_1 of nonempty strings over $\Sigma = \{a, b\}$ where the symbols in every third position are bs . So for example $aab, aabbbb \in L_1$. Use at most 4 states if you can. Submit this as the file **h3q3a.jff** to **Homework 3 Auto**.
 - The language L_2 of strings whose length is not two. Use at most 4 states if you can. Submit this as the file **h3q3b.jff** to **Homework 3 Auto**.
4. Design an NFA for each of the following languages over $\Sigma = \{a, b\}$ with at most the indicated number of states.
- The language L_3 of strings which contain at most three as , and all the as are consecutive. Use at most 4 states if you can. Submit this as the file **h3q4a.jff** to **Homework 3 Auto**.
 - The language L_4 of strings which contain abb or bba or aa as a substring. Use at most 6 states if you can. Submit this as the file **h3q4b.jff** to **Homework 3 Auto**.
5. Convert the following NFA into a DFA in which each state represents a subset of the states of the NFA. Please label the DFA states with a number containing the digits in the subset that state represents, in increasing order (for example represent the subset $\{1, 3, 2\}$ by the number 123):

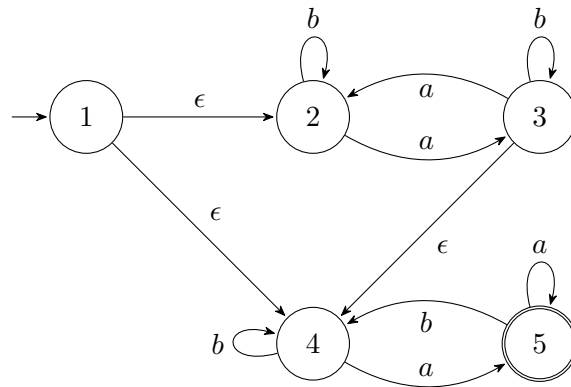


Figure 1:

Guidelines:

- Please submit your HW via Gradescope.
- Questions 3 and 4 must be submitted to the autograder at **Homework 3 Auto**.
- Questions 1, 2 and 5 must be handwritten then scanned to pdf and submitted to the autograder at **Homework 3 Pen**.
- If your handwriting is not legible, it will not be graded.
- Start the answer to each question on a separate page.

- You are allowed to collaborate with *others*, while doing your homework.
- Definition of *others*: Students who are taking the course this semester.
- You can consult GenAI tools.
- However, you should write the final solutions alone, without consulting anyone. Your writing should demonstrate that you understand the proofs completely.
- For each problem, you should acknowledge all the individuals you consulted and whether you used any GenAI resources. This will not affect your grade. Failure to acknowledge is considered *academic dishonesty*, and it will affect your grade.
- When proofs are required, you should make them both clear and rigorous. Do not hand wave.
- Any regrading requests should be made within one of returning the homework. No requests will be entertained after this one week period.